American Computer Science League

2023-2024 ● Contest 2: Short Problems ● Elementary Division Solutions

1. Prefix/Infix/Postfix 9 4 − 7 * 6 + is evaluated by looking for 2 numbers followed by an operator from left to right. Therefore, 9 4 − 7 * 6 + = (((9 4 −) 7 *) 6 +) = (5 * 7) + 6 = 41	C. 41
2. Prefix/Infix/Postfix Prefix expressions are evaluated by looking for 2 numbers preceded by an operator from left to right. Therefore, / + + * 2 4 6 * − 8 1 + 3 5 − 9 7 = / + + (* 2 4) 6 * (− 8 1) (+ 3 5) (− 97) = / + (+ (* 2 4) 6) (* (− 8 1) (+ 3 5)) (− 9 7) = / (+ ((+ (* 2 4) 6) (* (− 8 1) (+ 3 5)) (− 9 7) = (/ (+ ((+ (* 2 4) 6) (* (− 8 1) (+ 3 5)) (− 9 7)) = (/ (+ (+ 8 6) (* 7 8)) 2) = (/ (+ 14 56) 2) = (/ 70 2) = 35	C. 35
3. Prefix/Infix/Postfix First, rewrite this expression in infix using the order of operations: $\frac{8(7+5)-4^2*6}{9-2^3}$ ((8 * (7 + 5)) − ((4 ^ 2) * 6)) / (9 − (2 ^ 3)) = ((8 * (+ 7 5)) − ((^ 4 2) * 6)) / (− 9 (^ 2 3)) = ((* 8 (+ 7 5)) − (* (^ 4 2) 6)) / (− 9 (^ 2 3)) = (− (* 8 (+ 7 5)) (* (^ 4 2) 6)) / (− 9 (^ 2 3)) = / (− * 8 + 7 5 * ^ 4 2 6) (− 9 ^ 2 3) = / * − 8 + 7 5 * ^ 4 2 6 − 9 ^ 2 3	D. /−*8+75*^426−9^23

4. Prefix/Infix/Postfix To convert this expression to postfix, write each pair of numbers followed by the operator. Therefore, (7 * 2 ^ 3 − 9 * 6 + 8 / 4) ^ (7 − 2 * 3) = (7 * (2 3 ^) − (9 6 *) + (8 4 /)) ^ (7 − (2 3 *)) = ((7 (2 3 ^) *) − (9 6 *) + (8 4 /)) ^ (7 (2 3 * −)) = ((7 (2 3 ^) *) (9 6 *) −) + (8 4 /)) ^ (7 (2 3 * −)) = ((7 (2 3 ^) *) (9 6 *) −) (8 4 /) +) ^ (7 (2 3 * −)) = ((7 (2 3 ^) *) (9 6 *) −) (8 4 /) +) (7 (2 3 * −) ^) = 7 2 3 ^ * 9 6 * − 8 4 / + 7 2 3 * − ^	B. 723^*96*−84/+723*−^
5. Prefix/Infix/Postfix The expressions evaluate to: a. 4 3 + 2 * = (4 + 3) * 2 = 14 b. 4 3 2 + * = 4 * (3 + 2) = 20 c. + * 4 3 2 = (4 * 3) + 4 = 14 d. + 4 * 3 2 = 4 + (3 * 2) = 10 Therefore, both a and c have the same value.	E. a, c